DELIVERABLE 4.X - Accounting for spatiotemporal variation in life-history traits in stock assessment models

Rebecca Whitlock – SLU
Rob van Gemert - SLU

EMFAF-2023-PIA-FisheriesScientificAdvice
Project n° 101155914

Co-funded by the European Union
The EU is not responsible for the content of the project

Abstract

Survival during the first year at sea (or post-smolt-survival) is a key phase in the life-cycle of Atlantic salmon. It is modelled in different ways in stock assessments covering different parts of the species distribution, and in the Baltic Sea, the stock assessment model currently captures temporal, but not spatial variation in this parameter. The DIASPARA project tested alternative ways to capture spatiotemporal variation in survival during the first year at sea in the stock assessment model for Baltic salmon. Allowing spatial as well as temporal variation in post-smolt-survival led to large changes in estimated scale and development over time for some stocks, where more flexibility in the model structure generally allowed a closer fit to spawner counting data. Correlations among stocks in the temporal post-smolt-survival pattern were generally lower than expected and tended to be highest for stocks in the northern Baltic Sea. These results highlight the importance of accounting fir spatiotemporal variation in life-history traits in the stock assessment models, and form a starting point for improvements to the stock assessment model for Atlantic salmon in the Baltic Sea.

Index

Abstract 2

Index 3

1 Background 4

1.1 Stock assessment models for Atlantic salmon 5

2 Methods 9

2.1 Data 9

2.2 Changes implemented in the stock assessment model 10

3 Results 11

4 Discussion 24

5 References 25

1 Background

Accurate and biologically realistic stock assessment models are crucial for sustainable fisheries management. The level of harvest a stock can sustain depends on its productivity, which is shaped by a number of life-history traits including growth, maturation and survival (Beverton and Holt, 1957; Hilborn and Walters, 1992). Incorporating the effects that life history traits have on population dynamics is thus of high priority for stock assessment models. However, life history traits can vary both across time and space (Beukhof et al., 2019), making it challenging to accurately incorporate their effects in stock assessment models.

Incorporating variation in life history traits in stock assessment models is especially challenging for widely-distributed migratory species, such as Atlantic Salmon (*Salmo salar*) and European eel (*Anguilla anguilla*). Individuals of these species can make use of a wide variety of different habitats throughout their lives, which can lead to large differences in experienced environmental pressures over space and time, in turn leading to a wide variety of life history traits between individuals. Atlantic salmon experience their early life stages in freshwater river systems, where local conditions can differ significantly between different rivers and over time (Hutchings & Jones, 1998). Similarly, European eel experience a complex life cycle, from oceanic larval dispersal to continental freshwater or coastal growing habitats spanning most of continental Europe, with growth and maturation being strongly affected by local conditions across time and space (Durif et al., 2023). Given this large variety in habitat use over space and time for these species, it is crucial that their stock assessments incorporate how this affects their life history traits.

For the current ICES stock assessment of European eel, spatio-temporal variability in life history traits is not taken into account, as the current assessment only consists of an estimated index of continental glass eel recruitment (ICES, 2025a). National assessments are also made of local stock components, but these are made independent of each other, and do not feed into a larger 'global' stock assessment model. However, efforts are underway to develop regional assessment models that are meant to provide input for a larger global assessment model, incorporating the entirety of the European eel stock (ICES, 2021a; 2025a). For the further development of these models, it is crucial to better understand and incorporate the spatio-temporal variation of life history traits.

Two separate stock assessment models are currently used for Atlantic salmon; one covers the North Atlantic (ICES, 2025b), while the other covers the Baltic Sea (ICES, 2025c). These models share a similar approach to assessment – both are age- and stage-structured life-cycle models, with inference in a Bayesian framework. However, there are some key differences in model structure. For the North Atlantic, owing to the large number of salmon rivers there, stocks are assessed at the level of 25 stock units (each comprising multiple stocks), nested within three stock complexes (ICES, 2025b). The Baltic assessment, on the other hand, assesses stocks at the level of individual rivers. In addition, the Baltic assessment model includes a stock-recruitment function describing density-dependence during the freshwater part of the life-cycle, while the North Atlantic model does not currently account for density-dependence (although this is planned for the future). There are also differences in the level of detail with which life-history traits are

Co-funded by the European Union
The EU is not responsible for the content of the project

modelled, and the WGNAS model comprises a more comprehensive treatment of spatio-temporal variation in life-history traits (Table 1).

Table 1. Summary of how different life-history traits are modelled in ICES stock assessment models (SAM) for Atlantic salmon. WGNAS, Working Group for North Atlantic Salmon; WGBAST, Working Group for Baltic Salmon and Trout.

1.1 Stock assessment models for Atlantic salmon

Two separate stock assessment models are currently used for Atlantic salmon; one covers the North Atlantic (ICES, 2025b), while the other covers the Baltic Sea (ICES, 2025c). These models share a similar approach to assessment – both are age- and stage-structured life-cycle models, with inference in a Bayesian framework. However, there are some key differences in model structure. For the North Atlantic, owing to the large number of salmon rivers there, stocks are assessed at the level of 25 stock units (each comprising multiple stocks), nested within three stock complexes (ICES, 2025b). The Baltic assessment, on the other hand, assesses stocks at the level of individual rivers. In addition, the Baltic assessment model includes a stock-recruitment function describing density-dependence during the freshwater part of the life-cycle, while the North Atlantic model does not currently account for density-dependence (although this is planned for the future). There are also differences in the level of detail with which life-history traits are modelled, and the WGNAS model comprises a more comprehensive treatment of spatio-temporal variation in life-history traits (Table 1).

Table 1. Summary of how different life-history traits are modelled in ICES stock assessment models (SAM) for Atlantic salmon. WGNAS, Working Group for North Atlantic Salmon; WGBAST, Working Group for Baltic Salmon and Trout.

Life-history trait or transition	WGNAS SAM	WGBAST SAM
Egg-to-smolt	Density-independent egg-to-smolt survival, common for all stock units. Stock-unit specific smolt age distribution	Density-dependent stock-specific egg to smolt survival (Beverton-Holt model) One fixed smolt age for each river stock
Maturation	Proportion maturing as one sea-winter salmon. Varies in time (annually) and by stock unit	Proportion of immature fish maturing after one to five sea winters. Time varying (annual) with spring sea surface temperature as a covariate; not stock-specific

Co-funded by the European Union
The EU is not responsible for the content of the project

Fecundity	Fecundity by sea-winter age (eggs per female, including proportion of females). Time-varying but not yet stock unit specific.	Fecundity by sea winter age (eggs per female). Time invariant, not stock specific.
Post-smolt survival	Stock unit specific with covariation between stocks. Time-varying (annual).	Not stock specific. Time-varying (annual). Moving average structure

The stock assessment model (SAM) for Baltic salmon is an age- and stage- structured state-space hierarchical Bayesian population dynamics model (ICES, 2025c). It is a tag-integrated model that combines multiple data sets, including catch data, and river-specific data such as spawner counting and proportions of grilse and multi-sea-winter fish. In addition, a number of prior and likelihood approximations capture information from sub-models about stage-specific abundances (e.g. smolt abundances), and key population dynamics parameters (e.g. smolt carrying capacity).

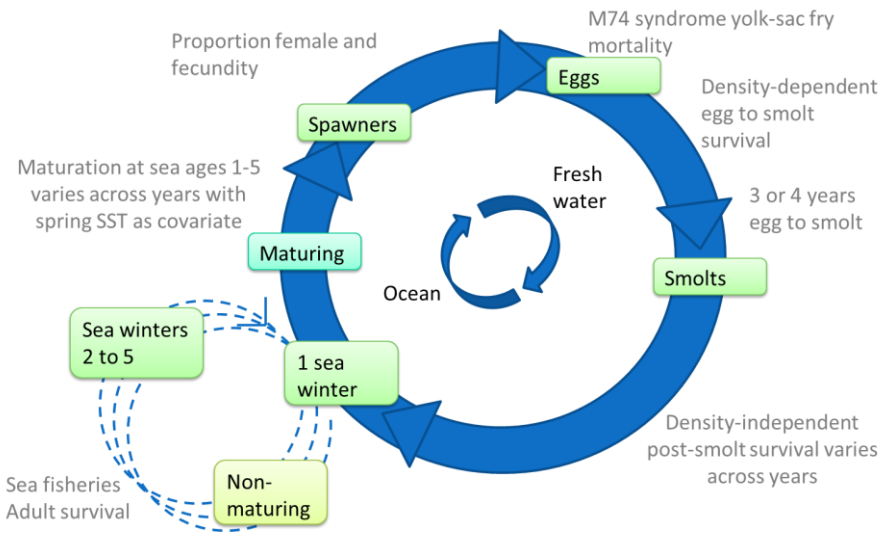

Figure 1. Schematic representation of the life-cycle model in the ICES WGBAST assessment model.

In the Baltic, wild salmon populations are assessed at the level of individual river stocks, and currently 17 wild stocks are included in the quantitative assessment. Individual river stocks are nested within assessment units (AUs) (Figure 1). The grouping of rivers within an assessment unit

Co-funded by the European Union
The EU is not responsible for the content of the project

is based on management objectives and biological and genetic characteristics of the river stocks. River stocks belonging to a particular unit are believed to exhibit similar migration patterns at sea, and are assumed to encounter the same sea fisheries and experience the same exploitation rates at sea. Baltic salmon stocks are managed according to the maximum sustainable yield (MSY) principle. This is implemented via calculation of stock-specific reference points for population size at MSY (R_{MSY}), and a limit reference point for population size (R_{lim}), corresponding to an abundance from which the MSY abundance can be reached in one generation time with no fishing, both expressed in numbers of smolts. R_{MSY} and R_{lim} reference points for Baltic salmon are calculated based on population dynamics at the unfished equilibrium, using a set of assumptions about future vital rates, including post-smolt survival (ICES 2025c, Table 4.3.1.1). Freeing up key life-history parameters to vary by stock has been on WGBAST's road map for model development for many years (ICES, 2021b). During recent years, it has become increasingly apparent that the current model structure fails to adequately capture differences in survival from smolt to spawner among stocks. This led to biases in estimates of stock size for certain stocks in 2024 and 2025 (ICES 2024, 2025c). A precautionary adjustment was made to the advised total allowable catch in 2025, based on concerns within the expert group that overestimation of abundance for the most numerous Baltic salmon stock (Tornionjoki) was contributing to overly-optimistic status evaluations for other wild Baltic salmon stocks (ICES, 2025c). Survival during the first year at sea (hereafter "post-smolt survival or PSS"), is a critical transition in the life-cycle of Atlantic salmon (e.g. Friedland et al. 2005), and has been shown to vary both through time and in space (Olmos et al. 2020). It is currently assumed to be equal for all wild salmon stocks in the WGBAST model. During the DIASPARA project, alternative model structures allowing spatiotemporal variation in post-smolt survival were evaluated.

Co-funded by the European Union
The EU is not responsible for the content of the project

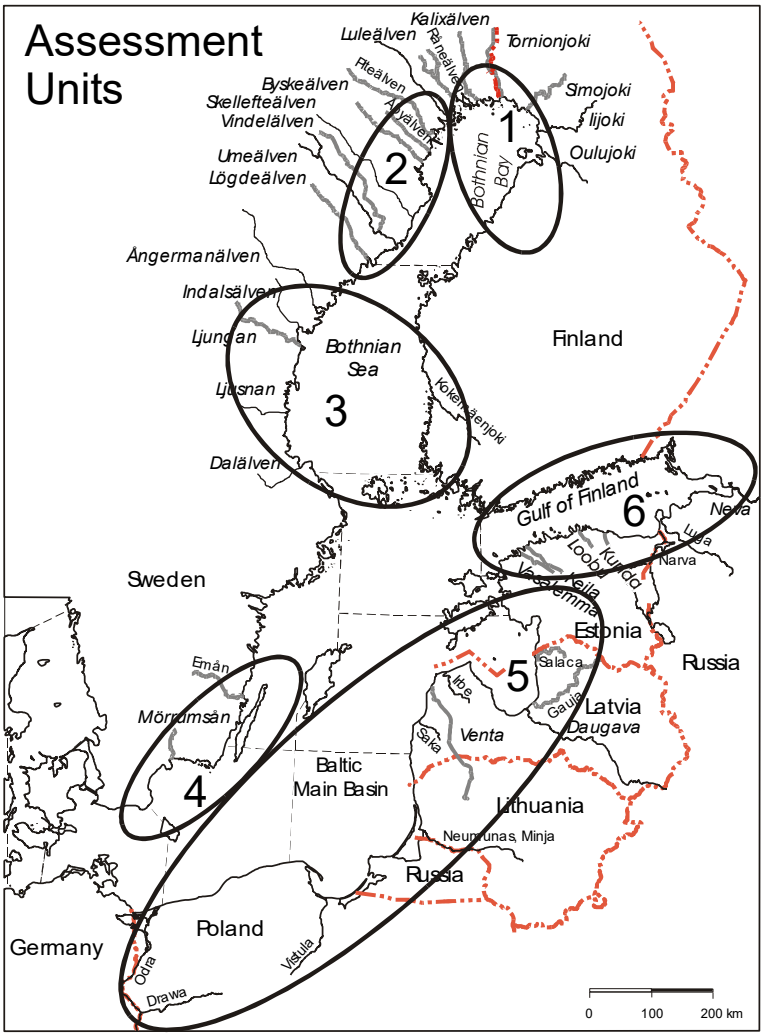

Figure 2. Map of the assessment units (AU) used by ICES WGBAST. Assessment units 1-4 are currently included in the quantitative assessment model. From ICES 2021c.

Co-funded by the European Union
The EU is not responsible for the content of the project

2 Methods

2.1 Data

Full information on data sets used in the assessment model can be found in the Stock Annex (ICES 2021c). Here, data sets with particular relevance to estimation of survival during the first year at sea are summarized. Rivers in the assessment model vary in the amount of available data, from rivers lacking both smolt and spawner counting, to rivers with good time series for smolt and spawners abundances (Table 2). In addition to smolt and spawner counting, tag recapture data between 1987 and 2012 (aggregated at the AU level) provide some information on rates of post-smolt survival. For rivers where no smolt counting data are available, time series of smolt abundance are obtained from a hierarchical Bayesian parr-to-smolt survival model that uses information on parr densities from electrofishing and estimates of production area together with smolt counting (where available) to produce time series of smolt abundance for all rivers in the assessment model (ICES, 2004, Annex 2). A hierarchical structure for the parr-to-smolt survival allows flow of information between rivers. Naturally, time series of smolt abundance for rivers that lack smolt counting carry much higher uncertainty.

Commented [RG1]: Which is this?

Commented [RW2R1]: ICES wgbast working group report 2004

Table 2. Summary of river-specific data in the WGBAST assessment model 2025 (model covers time span 1987-2024).

River	Assessment unit	Smolt counting (number of years, 1987-2024)	Spawner counting
Tornionjoki	1	27	2009-2024
Simojoki	1	36	2008-2024
Kalixälven	1	0	1990-2024
Råneälven	1	0	2014,2015,2022-2024
Piteälven	2	0	1987-2024
Åbyälven	2	2	-
Byskeälven	2	0	-
Kågeälven	2	0	-
Rickleån	2	3	-
Sävarån	2	8	-
Vindelälven	2	13	1987-2024
Öreälven	2	3	-
Lögdeälven	2	1	-
Ljungan	3	2	-
Testeboån	3	8	2016-2020,2022,2024
Emån	4	1	-
Mörrumsån	4	16	2022-2024

Commented [RW3]: Need to double check this

Co-funded by the European Union
The EU is not responsible for the content of the project

2.2 Changes implemented in the stock assessment model

The stock assessment model currently used by WGBAST uses an autoregressive model for post-smolt mortality, with a four-year moving average window (Equation 1). Post-smolt survival is assumed to be the same for all wild stocks, and post-smolt survival for reared stocks is assumed to be lower than, but proportional to survival for wild stocks over the assessment time frame.

1. $Mps_y \sim LogNormal(\log(\frac{Mps_{y-4:y-1}}{4}) - \frac{\sigma^2_{Mps}}{2}, \sigma^2_{Mps})$

2. $S_y = \exp(-Mps_y)$

In order to explore the implications for estimated stock development and status when spatio-temporal variation in life-history traits are accounted for, two alternative model structures for post-smolt survival have been implemented to date. The first uses the same structure that is currently implemented in the stock assessment model for North Atlantic salmon (ICES 2025b; Olmos et al. 2019), a multivariate random walk across years (Equation 3), with an estimated correlation matrix that describes the correlation between spatial units (either stocks or assessment units). Results from this model are presented here. For brevity, readers are referred to ICES 2021c for full details of the assessment model (codes for the models described here are available at https://github.com/DIASPARAproject/WP4_spatiotemporal_change_life_history_traits.

3. $logit(S_{y+1,1:R}) \sim MVNormal(S_{y,1:R}, \Sigma_{1:R,1:R})$

Where R denotes the number of spatial units (in this case 17 river stocks, or 4 assessment units).

4. $logit(S_{y,r}) = \beta_r + \delta_y + \alpha_{AU(r)} + \varepsilon_{y,r}$

5. $logit(S_{y,r}) = \beta_r + \delta_y + \alpha_{y,AU(r)} + \varepsilon_{y,r}$

An AR1 model is currently used to generate future scenarios for PSS, using the estimated time series from the model, and fitting a post-hoc autoregression analysis to estimate the stable mean, variance and correlation of the AR1 process. The stable mean to which the time series regresses is calculated based PSS estimates four years prior to the last year in the model up to the year prior to the last modelled year (i.e. 2020 to 2023 in 2025's assessment). In order to implement an equivalent process in the stock unit specific PSS scenarios, scenarios were generated using a multivariate AR1 model, using the stock-specific mean PSS on the logit scale for the four penultimate years, and using the estimate variance-covariance matrix for PSS. The autocorrelation parameter was calculated as the mean of stock-specific autocorrelation parameters from maximum likelihood fitting of AR1 processes for each PSS time series.

6. $logit(S_{y+1,1:R}) = \omega logit(S_{y,1:R}) + \gamma_{1:R},$

where

7. $\gamma_{1:R} \sim MVNormal(\mu_{1:R}(1-\omega), \Sigma_{1:R,1:R}(1-\omega^2)),$

Co-funded by the European Union
The EU is not responsible for the content of the project

ω is the average autocorrelation coefficient across stock units, $\mu_{1:R}$ is a vector of mean PSS on the logit scale from year $y-5$: $y-2$, and $\Sigma_{1:R,1:R}$ is the posterior median variance-covariance matrix for PSS.

Modified assessment models were run for 900,000 iterations, with a burn-in of 600,000 iterations and then thinned at an interval of 300 to yield 1000 samples per MCMC chain. Two MCMC chains were run in parallel, and convergence to the posterior distribution was assessed using Gelman-Rubin diagnostics.

3 Results

Convergence was assessed using visual inspection of trace plots and Gelman-Rubin diagnostics for monitored parameters. For the model with stock-specific PSS, 34% of Gelman-Rubin point statistics were below 1.10, and 62% were below 1.50, indicating that convergence for all parameters had not been reached. On closer inspection, one MCMC chain had become stuck at implausible values for some model parameters, and the results presented here are based on one MCMC chain. For the model with assessment unit specific PSS, 55% of Gelman-Rubin point statistics were below 1.10, and 83% were below 1.50, indicating better, but not perfect convergence to the posterior distribution. Results from this model are based on both MCMC chains. Since full convergence was not achieved in either model run, results presented below should be interpreted with caution.

Estimated time series for stock-specific post-smolt survival (PSS) showed large variation among stocks, with the majority experiencing decreasing survival over the assessment period (Figure 3). This decrease was most marked for stocks in AU 1, while stocks in AUs 2 and 3 had on average higher survival over the time series with a less pronounced decrease over time. The two stocks in AU 4 had low estimated post smolt survival over the whole time series. As expected, stocks with a larger amount of data displayed lower uncertainty in PSS estimates (Table 2, Figure 3). Pairwise correlations were modest (maximum (non-diagonal) of 0.34, between Kalixälven in AU 1, and Piteälven in AU 2), Figure 4, possibly indicating the lack of common drivers for PSS during the marine phase, and that local conditions may play a significant role. There was a pattern of decreasing correlation with decreasing latitude, so that some of the strongest correlations were found among the northernmost stocks (assessment unit one). Stronger correlations were also associated with data-rich rivers, for example Tornionjoki, Kalixälven and Piteälven, so that failure to identify correlations might also reflect lack of data. When PSS was estimated at the AU level, PSS showed a strong decrease over time for stocks in AUs 1 and 2 (Figure 5). PSS for AU 3 was on average higher than that for other AUs, and stable over time (Figure 5), while PSS for AU 4 stocks was low over the whole time series, and showed a decreasing trend. Aggregating stocks by AU generally led to stronger positive pairwise correlations between stock units, when compared with the stock-specific case. All assessment units showed positive correlations, with the exception of AU 3, which showed a negative correlation with AU 1 and was more or less uncorrelated with the temporal trend in AU 1 (Figure 6).

Co-funded by the European Union
The EU is not responsible for the content of the project

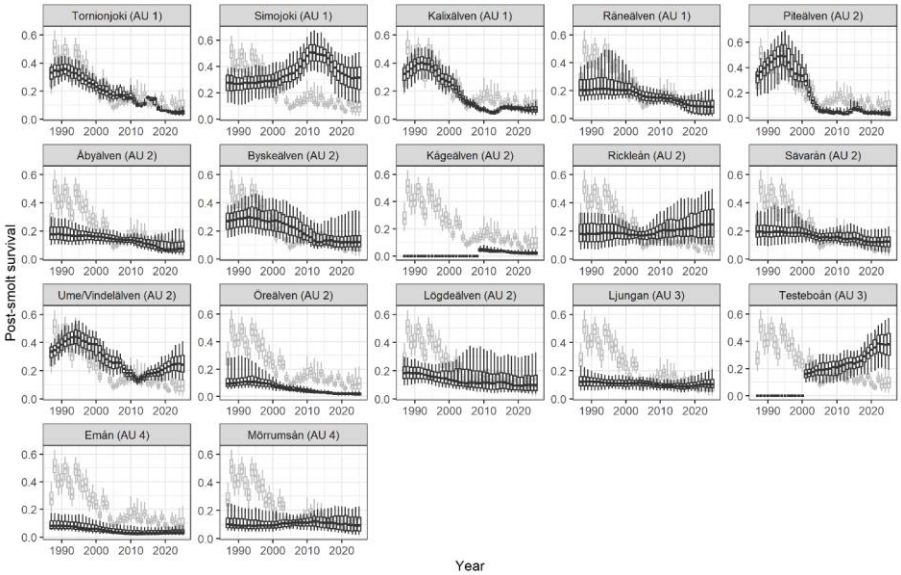

Figure 3. Boxplots of posterior distributions for post-smolt survival, 1987-2024. Horizonal lines within boxes denote medians, the ends of boxes denote the interquartile range, and whiskers denote the 90% posterior probability interval. Grey boxes, common post-smolt survival (2025 assessment model); black boxes, stock-specific post-smolt survival.

Co-funded by the European Union
The EU is not responsible for the content of the project

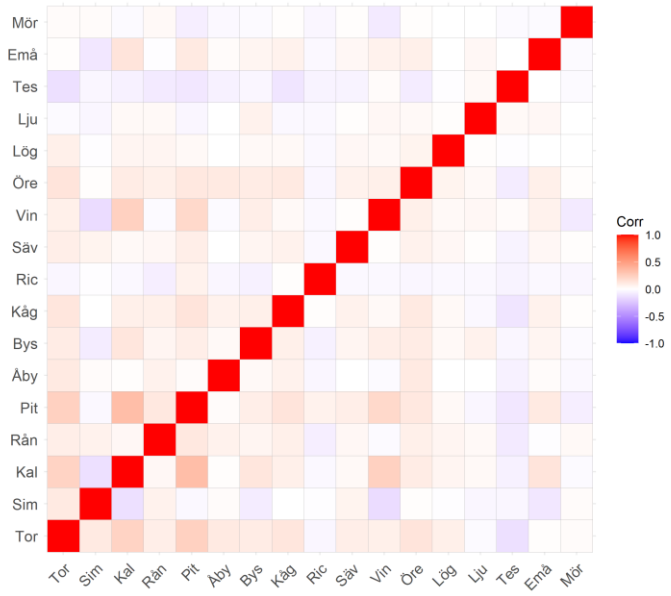

Figure 4. Correlation matrix showing posterior median pairwise correlations between 17 wild river stocks in the model with stock-specific post-smolt survival.

Co-funded by the European Union
The EU is not responsible for the content of the project

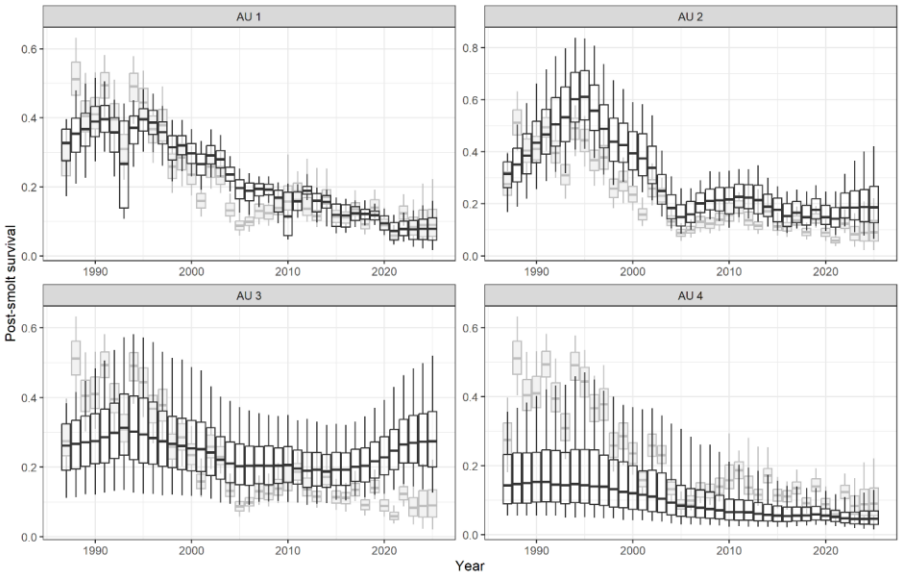

Figure 5. Boxplots of posterior distributions for post-smolt survival, 1987-2024. Horizonal lines within boxes denote medians, the ends of boxes denote the interquartile range, and whiskers denote the 90% posterior probability interval. Grey boxes, common post-smolt survival (2025 assessment model); black boxes, assessment unit (AU) specific post-smolt survival.

Co-funded by the European Union
The EU is not responsible for the content of the project

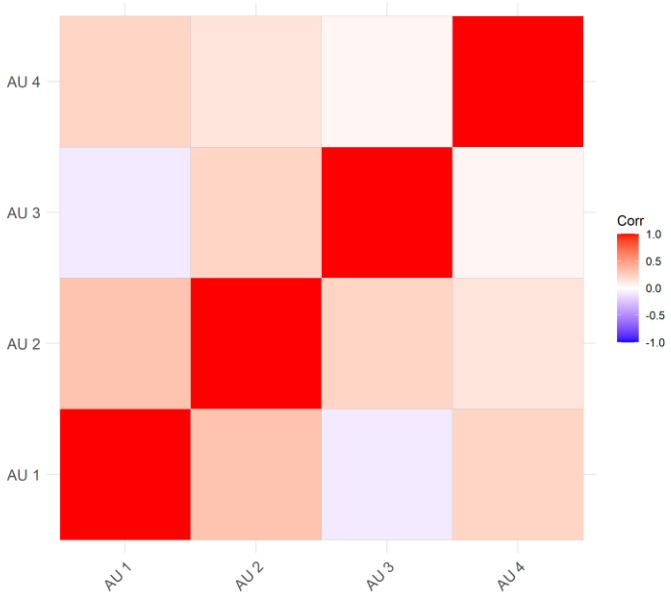

Figure 6. Correlation matrix showing posterior median pairwise correlations between four assessment units (AU) in the model with assessment unit specific post-smolt survival.

Estimating PSS by stock led to large updates in estimated spawner abundances, where estimates from the model with stock-specific PSS were often significantly different to those from 2025's assessment model with common PSS among stocks, based on non-overlapping 90% posterior probability intervals (Figures 7a and 7b). Allowing for PSS that varies by stock led to a much lower degree of updating to priors for detection probabilities for spawning salmon (results not shown), reflected in better adherence of spawner estimates to the temporal trend shown by spawner counting data (Figure 7a and 7b). Estimates of spawner abundance from the model where PSS was estimated by assessment unit were in many cases intermediate between those with stock-specific or common PSS (for example for Tornionjoki, Figures 7a and 8a). However, in some cases results from the model with stock-specific PSS were markedly different from those for common or assessment unit specific PSS, for example for Simojoki (Figures 7a and 8a), Kågeälven and Öreälven (Figures 7b and 8b). Allowing for stock-speciifc estimates of post-smolt survival led to estimates of unfished population size for certain stocks (for exmaple Kalixälven and Piteälven in Sweden), that were perceived as more realistic by experts.

Co-funded by the European Union
The EU is not responsible for the content of the project

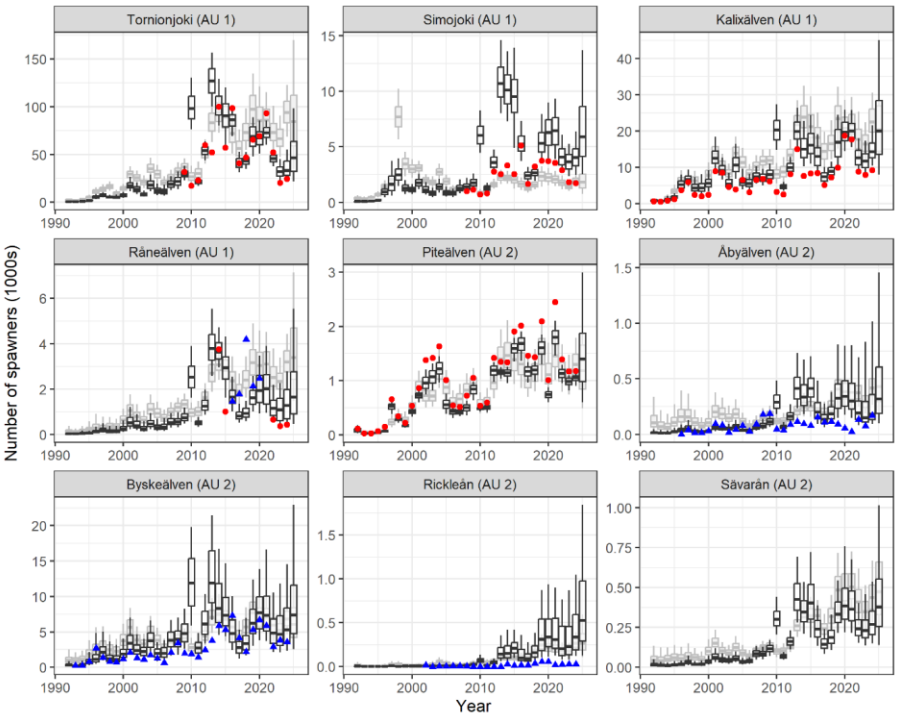

Figure 7a. Boxplots of posterior distributions for spawner abundance, 1992-2024. Horizonal lines within boxes denote medians, the ends of boxes denote the interquartile range, and whiskers denote the 90% posterior probability interval. Grey boxes, common post-smolt survival (2025 assessment model); black boxes, stock-specific post-smolt survival. Red circles indicate spawner counts that are used in model fitting; blue triangles indicate spawner counts that are not used in model fitting.

Co-funded by the European Union
The EU is not responsible for the content of the project

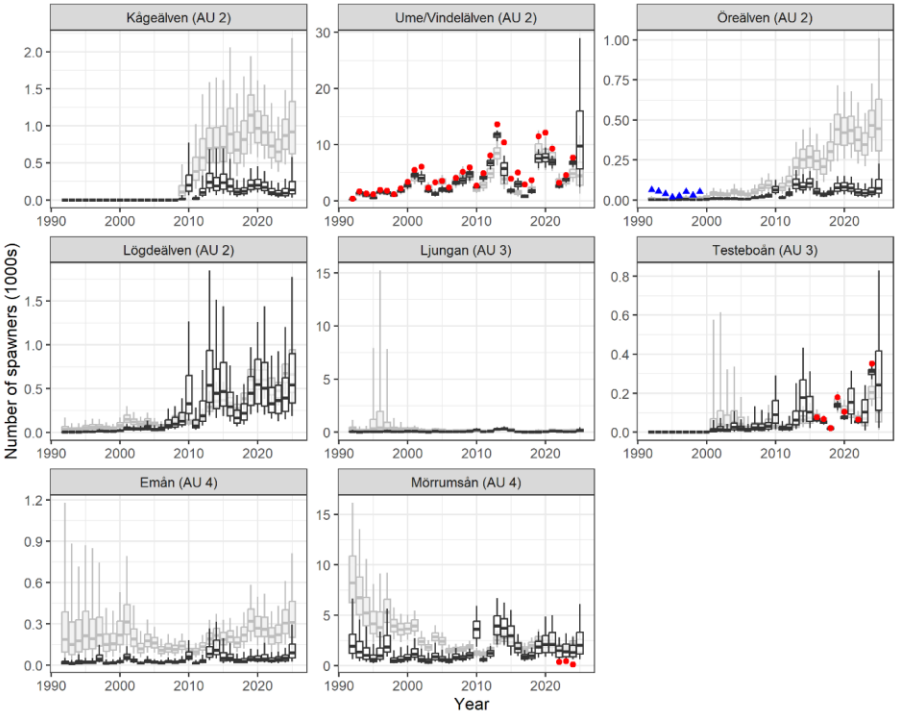

Figure 6b Boxplots of posterior distributions for spawner abundance, 1992-2024. Horizonal lines within boxes denote medians, the ends of boxes denote the interquartile range, and whiskers denote the 90% posterior probability interval. Grey boxes, common post-smolt survival (2025 assessment model); black boxes, stock-specific post-smolt survival. Red circles indicate spawner counts that are used in model fitting; blue triangles indicate spawner counts that are not used in model fitting.

Co-funded by the European Union
The EU is not responsible for the content of the project

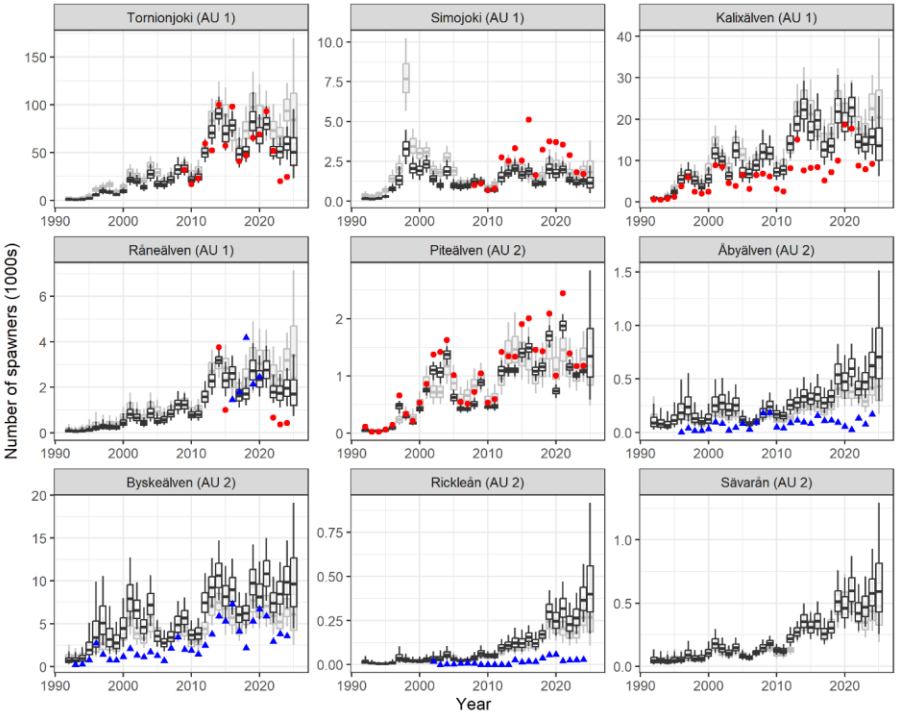

Figure 8a. Boxplots of posterior distributions for spawner abundance, 1992-2024. Horizonal lines within boxes denote medians, the ends of boxes denote the interquartile range, and whiskers denote the 90% posterior probability interval. Grey boxes, common post-smolt survival (2025 assessment model); black boxes, AU-specific post-smolt survival. Red circles indicate spawner counts that are used in model fitting; blue triangles indicate spawner counts that are not used in model fitting.

Co-funded by the European Union
The EU is not responsible for the content of the project

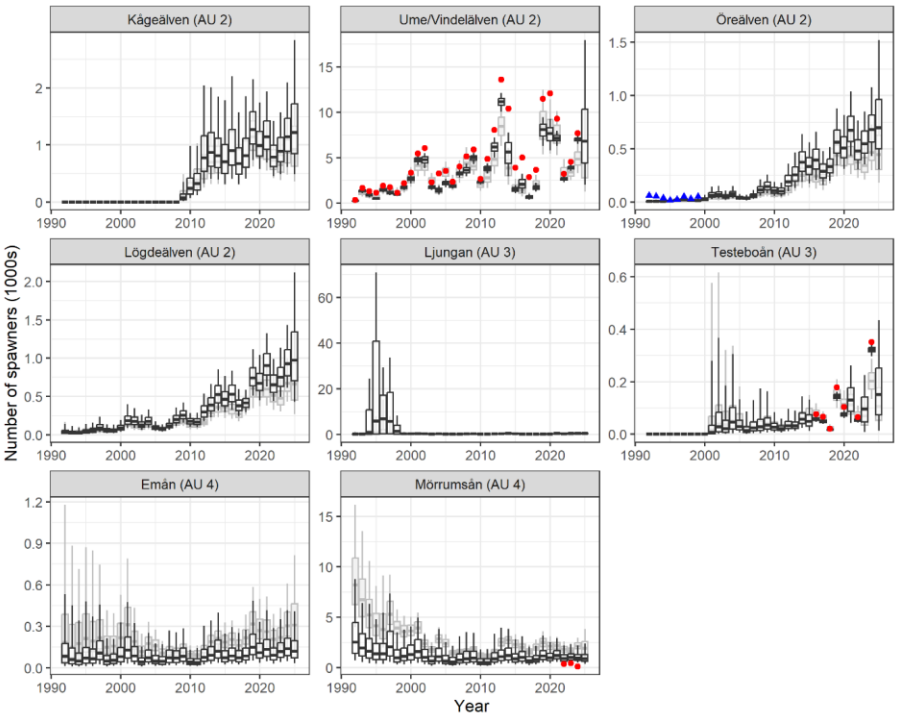

Figure 8b. Boxplots of posterior distributions for spawner abundance, 1992-2024. Horizonal lines within boxes denote medians, the ends of boxes denote the interquartile range, and whiskers denote the 90% posterior probability interval. Grey boxes, common post-smolt survival (2025 assessment model); black boxes, AU-specific post-smolt survival. Red circles indicate spawner counts that are used in model fitting; blue triangles indicate spawner counts that are not used in model fitting.

Using a different assumption about PSS also affected the estimated productivity of wild salmon stocks in the assessment model, via estimated stock-recruit steepness parameters (i.e. the proportion of the unfished recruitment expected when the spawning stock is reduced to 20% of its unfished level, note that the WGBAST model allows for steepness values less than 20% owing to the parameterization of the Beverton-Holt model used) (Figures 9 and 10). Changes in steepness were particularly marked for the model with stock-specific steepness, where higher steepness was estimated for many stocks (Figure 9), while a decrease in steepness was estimated for Piteälven. Estimated posteriors for stock-recruit steepness from the model with AU specific PSS generally overlapped with those from the 2025 assessment model with common PSS among stocks, with the exception of Testeboån in Sweden (Figure 10).

Estimated changes in steepness and smolt production at the unfished equilibrium (results not shown), were reflected in distributions for smolt abundance at MSY (Figures 11 and 12). These

Co-funded by the European Union
The EU is not responsible for the content of the project

overlapped with MSY distributions with common PSS to large degree for most stocks, for both PSS models. However there were some exceptions, particularly for the model with stock-specific PSS (for example Kalixälven and Piteälven. Note that the bimodal MSY distribution for Tornionjoki (AU 1) is likely related to non-convergence in the smolt carrying capacity for this stock (Figure 12). Changes in the reference points were reflected in changes in assessed stock status in 2024 (Table 3), where statuses for several stocks fell below the 50% threshold for probability to be above MSY population size used by the working group when estimating PSS by stock or assessment unit (Table 3).

Based on calculated WAIC values, the model with stock-specific PSS appeared to have greater support (WAIC lower by approximately 30 units) and also appeared to provide a closer fit to the spawner counting data for some rivers. However further runs and improvements in mixing and convergence are needed to more reliably compare WAIC values.

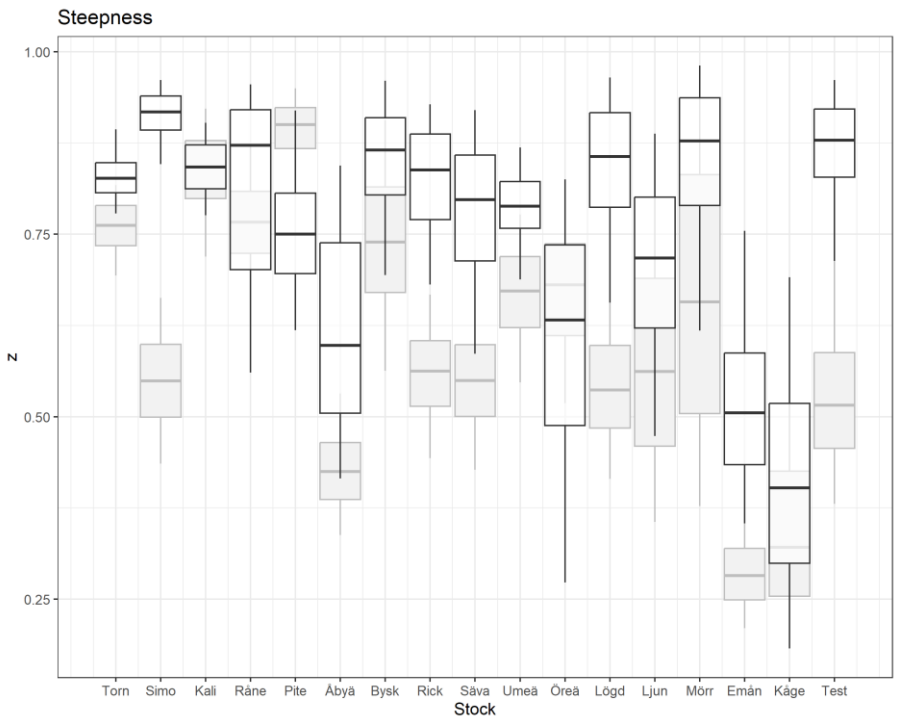

Figure 9. Posterior distributions for Beverton-Holt stock recuit steepness parameters. Grey boxes, 2025 assessment model; white boxes, stock-specific post-smolt survival.

Co-funded by the European Union
The EU is not responsible for the content of the project

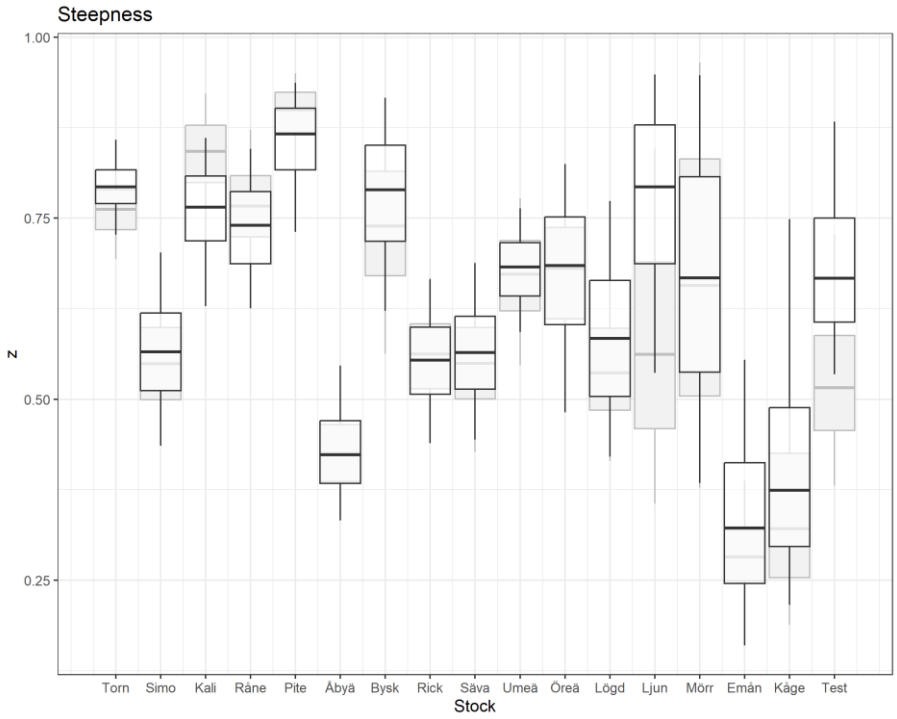

Figure 10. Posterior distributions for Beverton-Holt stock recuit steepness parameters. Grey boxes, 2025 assessment model; white boxes, assessment unit specific post-smolt survival.

Co-funded by the European Union
The EU is not responsible for the content of the project

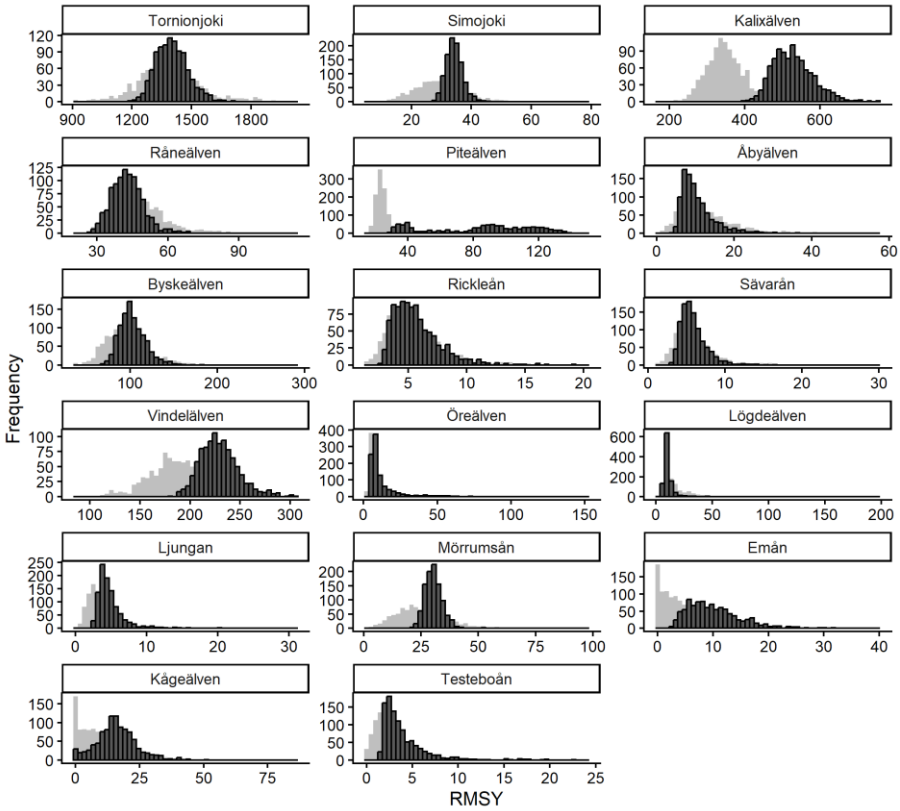

Figure 11. Histograms of posterior samples for the number of smolts at maximum sustainable yield (grey bars, 2025 assessment model; black bars, stock-specific PSS).

Co-funded by the European Union
The EU is not responsible for the content of the project

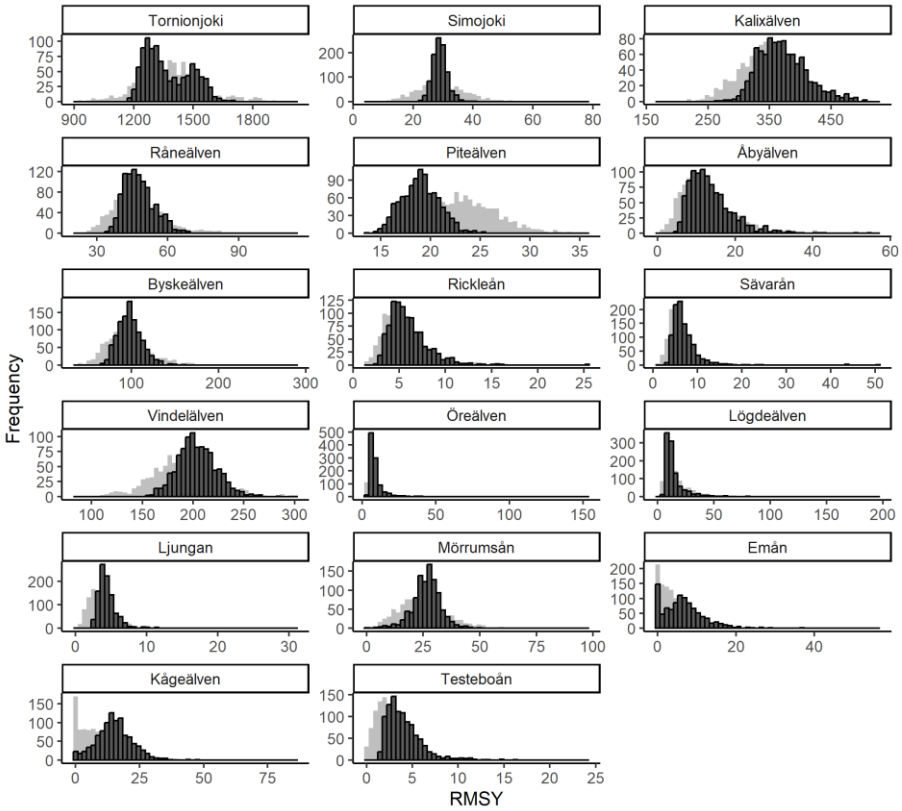

Figure 12. Histograms of posterior samples for the number of smolts at maximum sustainable yield (grey bars, 2025 assessment model; black bars, assessment unit specific PSS).

Table 3. Estimated stock status in 2024 (proportion of posterior samples for smolt abundance exceeding the stock-specific MSY reference point), for different model structures for PSS. Values below a 50% risk level threshold are highlighted in red, in accordance with current guidelines used by WGBAST.

Assessment unit	River	Probability to reach MSY in 2024, common PSS	Probability to reach MSY in 2024, stock-specific PSS	Probability to reach MSY in 2024, AU specific PSS

Co-funded by the European Union
The EU is not responsible for the content of the project

1	Tornionjoki	0.94	0.964	0.32
	Simojoki	0.88	0.906	0.995
	Kalixälven	0.93	0.806	0.91
	Råneälven	0.76	0.806	0.897
2	Piteälven	0.82	0.644	0.739
	Åbyälven	0.39	0.623	0.198
	Byskeälven	0.79	0.799	0.838
	Kågeälven	0.63	0.651	0.566
	Rickleån	0.60	0.447	0.413
	Sävarån	0.79	0.807	0.749
	Vindelälven	0.97	0.921	0.988
	Öreälven	0.65	0.569	0.662
	Lögdeälven	0.54	0.701	0.564
3	Ljungan	0.68	0.518	0.589
	Testeboån	0.70	0.445	0.451
4	Emån	0.50	0.184	0.39
	Mörrumsån	0.73	0.779	0.861

4 Discussion

Results from models that allow spatiotemporal variation in PSS demonstrate clear variations between populations, underlining the importance of accounting for spatiotemporal variation in this parameter to increase the biological realism and accuracy of the stock assessment model. Failure to account for stock-specific differences in the current assessment is thus likely contributing to poor predictions of spawners returns for some stocks in recent years (ICES 2025c). We found a pattern of higher synchrony in PSS in the more northerly populations in the Baltic Sea, possibly indicating underlying common drivers (and/or possibly more data-rich rivers in the region). Baltic salmon experts found the the results from the model with stock-specific PSS more plausible than those where PSS varies by AU for some rivers, e.g. Piteälven, however, this model was thought to underestimate spawner abundance for some rivers (Åbyälven, Rickleån). This can likely be improved by including spawner counting data for as many rivers as available in future model runs.

Results from this study highlight the importance of choosing a suitable model structure for PSS. The underlying model structure can cause large differences in estimated population scale and productivity, leading to changes in estimated status relative to management reference points (Table 3). Since the river stock with the poorest status generally drives TAC setting for Baltic

Co-funded by the European Union
The EU is not responsible for the content of the project

salmon, choice of model structure for PSS has important implications for management. A review of the model structure is currently underway, and in it, alternative model structures will be tested, for example methods that partition the total spatiotemporal variation into trends at different levels of aggregation (Grosbois et al. 2009; Olmos et al. 2020). Furthermore, an external study will be conducted to assess whether there are any covariates (with a sound mechanistic rationale) that could be included to improve short term predictions of abundances of salmon returning to spawn.

The current benchmark process for Baltic salmon (in which the model will be developed and reviewed), will also include expansion of the model to include rivers in AU 5. Several aspects should thus be considered when selecting a model structure to move forward with the SAM for Baltic salmon, including the following:

- which underlying model structure for PSS is most plausible from a biological/ecological perspective
- ease of adding environmental covariates into the new PSS structure
- predictive distribution for a new stock unit, for example a new stock within an existing, or new assessment unit (i.e. before observing any data from that stock unit)
- plausibility of PSS estimates for rivers whose time series starts later than others in the model (i.e. after 1987) (Kågeälven, Testeboån).
- posterior model probabilities/likelihood-based information criteria to measure probability of alternative model structures – Bayes Factors, WAIC etc.
- MCMC mixing and run time to achieve an acceptable level of convergence

Poor mixing and long run times (weeks) have been an ongoing problem for fitting of the WGBAST assessment model. Recent work to simplify the process error structure for survival parameters has resulted in an approximate halving of model run time with no noticeable impact on results, however, further progress is needed. Results from task 4.2 of DIASPARA show promise for customizing MCMC samplers in Nimble to improve mixing for this model. Further simplifications of the model and regularization of priors (Monnahan et al. 2019) will likely also be needed.

Commented [RW4]: is this the right one?

Commented [RG5R4]: Yup!

5 References

Beukhof, E., Dencker, T. S., Pecuchet, L., & Lindegren, M. (2019). Spatio-temporal variation in marine fish traits reveals community-wide responses to environmental change. Marine Ecology Progress Series, 610, 205-222.

Beverton, R. J., & Holt, S. J. (1957). On the dynamics of exploited fish populations. Great Britain Ministry of Agriculture, Fisheries and Food, London.

Durif, C. M. F., Arts, M., Bertolini, F., Cresci, A., Daverat, F., Karlsbakk, E., ... & Browman, H. I. (2023). The evolving story of catadromy in the European eel (Anguilla anguilla). ICES Journal of Marine Science, 80(9), 2253-2265.

Friedland, K. D., Chaput, G., MacLean, J. G. The emerging role of climate in post-smolt growth of Atlantic salmon, ICES Journal of Marine Science, Volume 62, Issue 7, 2005, Pages 1338–1349, https://doi.org/10.1016/j.icesjms.2005.04.013

Co-funded by the European Union
The EU is not responsible for the content of the project

Grosbois, V., Harris, M. P., Anker-Nilssen, T., McCleery, R. H., Shaw, D. N., Morgan, B. J., & Gimenez, O. (2009). Modeling survival at multi-population scales using mark–recapture data. Ecology, 90(10), 2922-2932.

Hilborn, R., & Walters, C. J. (1992). Quantitative fisheries stock assessment: choice, dynamics and uncertainty. Chapman and Hall., Inc.

Hutchings, J. A., & Jones, M. E. (1998). Life history variation and growth rate thresholds for maturity in Atlantic salmon, Salmo salar. Canadian Journal of Fisheries and Aquatic Sciences, 55(S1), 22-47.

ICES. (2021a). Workshop on the future of eel advice (WKFEA). ICES Scientific Reports. 3:13. 67 pp. https://doi.org/10.17895/ices.pub.5988

ICES. (2021b). Baltic Salmon and Trout Assessment Working Group (WGBAST). ICES Scientific Reports. 3:26. 331 pp. https://doi.org/10.17895/ices.pub.7925

ICES. (2021c). Stock Annex: Salmon (Salmo salar) in subdivisions 22–31 (Main Basin and Gulf of Bothnia) and Subdivision 32 (Gulf of Finland). ICES Stock Annexes. Report. https://doi.org/10.17895/ices.pub.18623147.v1

ICES. (2024). Baltic Salmon and Trout Assessment Working Group (WGBAST). ICES Scientific Reports. 6:42. 425 pp. http://doi.org/10.17895/ices.pub.25868665

ICES. (2025a). Joint EIFAAC/ICES/GFCM Working Group on Eels (WGEEL). ICES Scientific Reports. 7:99. 134 pp. https://doi.org/10.17895/ices.pub.30488120

ICES. (2025b). Working Group on North Atlantic Salmon (WGNAS). ICES Scientific Reports. 7:44. 435 pp. https://doi.org/10.17895/ices.pub.28777226

ICES. (2025c). Baltic Salmon and Trout Assessment Working Group (WGBAST). ICES Scientific Reports. 7:50. 378 pp. https://doi.org/10.17895/ices.pub.29118545

Monnahan, C. C., Branch T. A., Thorson, J. T., Stewart, I. J., Szuwalski, C. S. Overcoming long Bayesian run times in integrated fisheries stock assessments, ICES Journal of Marine Science, Volume 76, Issue 6, November-December 2019, Pages 1477–1488,

Olmos, M., Massiot-Granier, F., Prévost, E., Chaput, G., Bradbury, I. R., Nevoux, M., & Rivot, E. (2019). Evidence for spatial coherence in time trends of marine life history traits of Atlantic salmon in the North Atlantic. Fish and Fisheries, 20(2), 322-342.

Olmos, M., Payne, M. R., Nevoux, M., Prévost, E., Chaput, G., Du Pontavice, H., ... & Rivot, E. (2020). Spatial synchrony in the response of a long range migratory species (Salmo salar) to climate change in the North Atlantic Ocean. Global Change Biology, 26(3), 1319-1337.

Co-funded by the European Union
The EU is not responsible for the content of the project